

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of BSc (Biochemistry) Examination April 2019

Course code & Name BC406: Enzymology

Date: 03-05-2019 Day: Friday Time: 2:00 PM To 4:30 PM Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEETS.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I											
Q - II Answer the following questions as directed		20									
Q- II (A)	Answer any FOUR of the following (i) Explain allosteric enzyme regulation and kinetics in detail with an example. (ii) Describe Uncoupled and coupled spectrophotometric method of enzyme assays with example. (iii) In Lab practical's, a student extracted total protein from overnight grown bacteria. Now he wants to separate the 100kDa, 50 kDa and 10 kDa protein from the total protein. Suggest and explain the most suitable method for the given experiment. (iv) Explain any two enzymes of significance in disease diagnosis. (v) Solve the following problems- (a) The slope and intercept obtained from (1/rate) against (1/ substrate conc.) of an enzyme catalyzed reaction are 300 and 2×10^5 respectively. Find out the K_m of the enzyme in this reaction. (b) A company has developed 2 enzymes that degrade highly toxic compounds to nontoxic compounds. Your task is to degrade the greatest amount of the toxic compounds in the shortest amount of time. The kinetic properties of two enzymes are shown below- <table><tr><th>Enzyme</th><th>K_m</th><th>V_{max}</th></tr><tr><td>Enzyme 1</td><td>2mM</td><td>100mM/sec</td></tr><tr><td>Enzyme 2</td><td>10mM</td><td>500mM/sec</td></tr></table> At <u>saturating concentrations</u> of substrate, which enzyme would be better to use? Why?	Enzyme	K_m	V_{max}	Enzyme 1	2mM	100mM/sec	Enzyme 2	10mM	500mM/sec	20
Enzyme	K_m	V_{max}									
Enzyme 1	2mM	100mM/sec									
Enzyme 2	10mM	500mM/sec									
SECTION - II											
Q-III Answer the following questions as directed		30									
Q- III (A)	Answer any ONE of the following (i) Draw the Line weaver Burk plots for- competitive, noncompetitive and	5									

	uncompetitive inhibition. Mention the effect on K_m and V_{max} in the presence of these inhibitors. (ii) What is rate of chemical reaction? Explain the factors affecting the rate of chemical reaction.	
Q- III (B)	Answer any FIVE of the following (i) Derive Michaelis menton equation. (ii) Define Isoenzyme and Zymogen with examples. (iii) Define the role of the following compounds during enzyme extraction a) Lytic enzymes/detergents b) Protease and Phosphatase inhibitors (iv) Explain Hyperbolic graph of enzyme kinetics. (v) Explain any enzyme (protein) purification method based on solubility. (vi) Define K_{cat} and V_{max}.	15
Q- III (C)	Answer any FIVE of the following (i) Discuss about different matrix used in immobilization technology. (ii) Define a) Katal b) 1 unit specific activity (iii) Short note on Oligomeric enzymes with an example. (iv) Short note on Transition state theory. (v) What is enzyme commission number and mention the different classes of enzymes. (vi) Brief note on Ribozymes. Give some examples.	10